# Maths for Computing — Phase 2: Sets and Intervals

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## What this phase covers

This phase is all of Lecture 2 and Seminar 2, the biggest chunk of your exam. It unlocks mock **Questions 1, 8, 9, 10, 11, 13 and 14**, plus revision Tasks 4, 5, 12 and 13.

You need from earlier phases: inequality signs (Phase 0), squares and square roots (Phase 0), and mod (Phase 1).

Keep the Phase 0 symbol dictionary open while you read. Sets use many symbols, and each one is explained again here the first time it appears.

## 1. What a set is

A **set** is a collection of **distinct** (all different) objects. The objects in a set are called its **elements** or **members**.

- Sets are written with **curly braces** { }, with the elements separated by commas: {1, 2, 3, 5, 7}.
- Elements can be anything: numbers, letters, symbols, words, even **other sets**. {☼, 5, ♣} is a valid set.
- Sets are usually named with capital letters: A = {a, b, c}.

### Two key rules of sets

1. **Order does not matter.** {1, 2, 3} and {3, 1, 2} are the same set.
2. **Repeats do not count.** {1, 1, 2} is the same set as {1, 2}. Each element is either in the set or not; it can't be "in twice".

### Membership: ∈ and ∉

- **x ∈ A** means "x is an element of A", or simply "x is in A". Example: 2 ∈ {1, 2, 3}.
- **x ∉ A** means "x is not in A". Example: 5 ∉ {1, 2, 3}.

The symbol ∈ looks like a curly E, for "element".

## Sets versus sequences

Phase 1 had sequences. They look similar but follow opposite rules.

| Property | Set | Sequence |
|---|---|---|
| Written with | Curly braces { } | Round brackets ( ) or angle brackets ⟨ ⟩ |
| Order | Does not matter | Matters |
| Repetition | Not allowed (repeats ignored) | Allowed |
| Positions | No positions (not indexed) | Each term has an index: 1st, 2nd, … |
| Size | **Cardinality**: number of distinct elements | **Length**: number of terms, including repeats |

Example: the sequence (1, 1, 2) has length 3. The set {1, 1, 2} = {1, 2} has cardinality 2.

## Famous sets

From Phase 1: $\mathbb{N}$ (natural numbers), $\mathbb{Z}$ (integers), $\mathbb{Q}$ (rationals), $\mathbb{R}$ (reals). These are **infinite** sets: they never end.

## Two special sets

- The **empty set**, written $\varnothing$ or **{ }**, has no elements at all. Think of an empty folder.
- The **universal set**, written **U**, contains every element being considered in the problem. Example: if a problem is about customers 1 to 7, then U = {1, 2, 3, 4, 5, 6, 7}. U changes from problem to problem; the question tells you what it is.

## Mini tasks 1

1. Write the set of vowels in the English alphabet.
2. Write the set of months with 31 days.
3. True or false: {a, b, c} = {c, a, b}.
4. True or false: $3 \in \{1, 2, 3\}$. And: $\{3\} \in \{1, 2, 3\}$?
5. From A = {2, 3, 4, 5, 6, 7, 8, 9}, write the set of elements that are prime.

# 2. Cardinality: how many elements?

The **cardinality** of a set A is the number of distinct elements in it. It is written **|A|**.

- |{a, b, c}| = 3
- |{−5, 0, 5, 10}| = 4
- |{ }| = |∅| = **0**

## The golden rule: count only the top-level commas

When a set contains other sets, each inner set counts as **one** element, however much is inside it. Think of a box: a smaller box inside counts as one item, even if it is full.

**Method:**

1. Strip off the outer braces.
2. Split what remains at the commas that are **not inside any inner braces**.
3. Remove duplicates.
4. Count.

## Worked examples (lecture activity)

**C = {{a, {b}}, {c, d}}**

- Strip the outer braces: {a, {b}}, {c, d}.
- Top-level pieces: {a, {b}} and {c, d}.
- **|C| = 2.**

**D = {{ }, { }, 10, 11}**

- Pieces: { }, { }, 10, 11.
- The two empty sets are the same element, so one is a duplicate.
- Distinct elements: { }, 10, 11. **|D| = 3.**

**E = { }** is empty: **|E| = 0**.

So |D| + |E| − |A| with A = {α, β, γ} gives 3 + 0 − 3 = 0.

## Important: ∅ versus {∅}

- ∅ = { } is an empty box. It has **0** elements.
- {∅} = {{ }} is a box containing an empty box. It has **1** element.

The lecture says: if empty sets are included on purpose as elements, they count as elements.

### More examples (Seminar 2 Tasks 1.3 and 1.4)

**K = {{H, B, {M}}, {I}, {J}}** — pieces: {H, B, {M}}, {I}, {J}. **|K| = 3**. (It doesn't matter that H or J were names of other sets; inside K they are just symbols.)

**F = {red, blue, yellow, { }}** — pieces: red, blue, yellow, { }. **|F| = 4.**

**A = {{a, b}, { }, {c}, {d, e, {f}}}** — four pieces. **|A| = 4.**

**B = {1, 2, 3, { }, { }}** — pieces 1, 2, 3, { }, { }. The two empty sets are the same. **|B| = 4.**

**C = {{ }, { }, {{ }, { }}}** — pieces { }, { }, {{ }, { }}. The first two are duplicates. So distinct elements: { } and {{ }, { }}. **|C| = 2.** (Also note {{ }, { }} = {{ }}, since repeats inside collapse too.)

### Mini tasks 2

1. |{x, y, z, w}|
2. |{1, {1}, {1, 2}}|
3. |{{ }}|
4. |{a, a, b, {a, b}}|
5. |{{1, 2}, {2, 1}, 3}| (careful!)

## 3. Equal sets, subsets and proper subsets

### Equal sets

Two sets are **equal** (A = B; the lecture also writes A ≡ B) when:

1. every element of A is in B, **and**
2. every element of B is in A.

So {1, 2, 3} = {3, 2, 1} and {♣, ☼, 5} = {☼, 5, ♣}. Order is ignored.

### Subset: ⊆ ("is a subset of", like ⩽)

A is a **subset** of B, written **A ⊆ B**, if **every** element of A is also in B. B may have extra elements; A may not.

- {1, 2} ⊆ {1, 2, 3, 4}: yes, 1 and 2 are both in the bigger set.
- {1, 5} ⊆ {1, 2, 3, 4}: no, 5 is missing.
- {1, 2, 3, 4} ⊆ {1, 2, 3, 4}: **yes**. Every set is a subset of itself.

The lecture calls ⊆ a **weak** subset and compares it to "less than or equal to" (⩽).

Two facts you must memorise:

- **The empty set is a subset of every set.** ∅ ⊆ A always. (It has no elements, so none of them can be missing from A.)
- **A ⊆ A** for any set A.

## Proper subset: ⊂ ("is a proper subset of", like <)

A is a **proper subset** of B, written **A ⊂ B**, if A ⊆ B **and** A ≠ B. So B must have at least one element that A lacks.

The lecture calls this a **strong** subset and compares it to "less than" (<).

- {a, b} ⊂ {a, b, c, d, e}: yes. c, d, e are extra.
- {a, b, c, d, e} ⊂ {a, b, c, d, e}: **no**. They are equal. A set is never a proper subset of itself.
- But {a, b, c, d, e} ⊆ {a, b, c, d, e}: yes.

## Don't mix up ∈ and ⊆

This is a classic exam trap.

- **∈** links an **element** to a set: 1 ∈ {1, 2}.
- **⊆** links a **set** to a set: {1} ⊆ {1, 2}.

So:

| Statement | True? | Why |
|---|---|---|
| 1 ∈ {1, 2} | True | 1 is an element |
| {1} ∈ {1, 2} | False | The set {1} is not listed as an element |
| {1} ⊆ {1, 2} | True | Every element of {1} is in {1, 2} |
| 1 ⊆ {1, 2} | Wrong form | 1 is not a set |
| ∅ ⊆ {1, 2} | True | The empty set is a subset of everything |
| ∅ ∈ {1, 2} | False | ∅ isn't listed as an element |

| Statement | True? | Why |
|---|---|---|
| $\{1\} \in \{\{1\}, 2\}$ | True | Here $\{1\}$ **is** listed as an element |

### Mini tasks 3

True or false?

1. $\{1, 2, 3\} \subseteq \{3, 2, 1\}$
2. $\{1, 2, 3\} \subset \{1, 2, 3\}$
3. $\varnothing \subset \{a\}$
4. $\{a, d\} \subseteq \{a, b, c\}$
5. $\{2\} \in \{1, 2, 3\}$

## 4. Set builder notation

Listing every element is not always possible (or practical). **Set builder notation** describes a set by the **rule** its elements must follow.

```
\{\, x \mid \text{condition on } x \,\}
```

- **x** is a placeholder variable: "a typical element".
- The vertical bar **|** (or a colon **:**) is read **"such that"**.
- After the bar comes the **condition**. Several conditions can be joined with "and", a comma or a semicolon; all must be true.

Read $\{x \mid x \in A; x > 3\}$ as: "the set of all x such that x is in A and x is greater than 3."

The form $\{x \in A : \text{condition}\}$ means the same as $\{x \mid x \in A \text{ and condition}\}$: x is taken from A, and must also satisfy the condition.

### How to read and solve one

1. Find **where x comes from** (the set after $\in$). That gives the candidates.
2. Test **each candidate** against the condition.
3. Keep the ones that pass.

## Examples from the lecture

- A = {2, 4, 6, 8, 10}. {x | x ∈ A; x mod 4 = 0} → test each: 2 no, 4 yes, 6 no, 8 yes, 10 no. Result **{4, 8}**.
- {x | x ∈ {a, b, c, d, e} and x < d} → letters before d alphabetically: **{a, b, c}**.
- {x | x ∈ {1, 2, 3, 4, 5}; x > 3} → **{4, 5}**.

## Examples from Seminar 2 Task 1.2

A = {2, 3, 4, 5, 6, 7, 8, 9}.

- Elements x where 2x is also in A: 2 → 4 ✓, 3 → 6 ✓, 4 → 8 ✓, 5 → 10 ✗, and the rest are too big. Result **{2, 3, 4}**.
- Elements x where $x^2$ is also in A: 2 → 4 ✓, 3 → 9 ✓, 4 → 16 ✗. Result **{2, 3}**.

## Conditions that are equations (mock Question 10)

**$\{x \in \mathbb{Z} : x^3 = -8\}$** means "all integers whose cube is −8". Solve the equation: x = −2 is the only number with $(-2)^3 = -8$. Is −2 an integer? Yes. So the set is **{−2}**. Answer **B**.

Why not {−2, 2}? Because $2^3 = +8$, not −8. Cubes keep the sign. (Compare $\{x \in \mathbb{Z} : x^2 = 4\} = \{-2, 2\}$, because squaring loses the sign.)

**The set x comes from matters.**

- $\{x \in \mathbb{Z} : x^2 = 2\} = \emptyset$, because $\sqrt{2}$ isn't an integer.
- $\{x \in \mathbb{R} : x^2 = 2\} = \{-\sqrt{2}, \sqrt{2}\}$.
- $\{x \in \mathbb{N} : x^2 = 2x\}$: $x^2 - 2x = 0$, so $x(x - 2) = 0$, giving x = 0 or x = 2. If $\mathbb{N} = \{1, 2, 3, \ldots\}$, only **{2}**.
- $\{x \in \mathbb{N} : x^2 = x\}$: x = 0 or 1. With $\mathbb{N}$ starting at 1, **{1}**.

## Sets of pairs (tuples)

An **ordered pair** or **tuple** (x, y) is two values in a fixed order. Sets can contain tuples.

**Lecture Activity 2:** A = {1, 2, 3}, B = {a, b}.

- S = {(x, y) | x ∈ A, y ∈ B} = {(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)}.
- T = {(x, y) | x ∈ A, y ∈ B, x > 2} = {(3, a), (3, b)}.

**Seminar Task 4.1(c):** {(x, y) | x ∈ ℕ, y ∈ ℕ, x < y and y ⩽ 3}, with ℕ = {1, 2, 3, ...}.

- y can be 1, 2 or 3. For each y, x must be a natural number smaller than y.

- y = 1: no x (x would have to be below 1).
- y = 2: x = 1 → (1, 2).
- y = 3: x = 1 or 2 → (1, 3), (2, 3).
- Result {(1, 2), (1, 3), (2, 3)}, cardinality **3**.

### Mini tasks 4

1. List $\{x \mid x \in \{1, 2, ..., 10\}, x \text{ is odd and } x > 4\}$.
2. List $\{x \in \mathbb{Z} : x^2 < 5\}$.
3. List $\{x \in \mathbb{Z} : x^3 = 27\}$.
4. List $\{(x, y) \mid x \in \mathbb{N}, y \in \mathbb{N}, x = y \text{ and } y < 5\}$, with $\mathbb{N} = \{1, 2, 3, ...\}$.
5. Write "the set of even natural numbers less than 10" in set builder notation, then list it.

## 5. Set operations and Venn diagrams

An **operation** on sets takes sets and produces a new set. There are four, all from the lecture. Running example: **A = {1, 2, 3, 4}, B = {3, 4, 5, 6}, C = {6, 7, 8}, D = { }**.

### Union: A ∪ B ("A or B")

Everything that is in A **or** in B (or both). Combine and remove duplicates.

- A ∪ B = {1, 2, 3, 4, 5, 6}
- B ∪ C = {3, 4, 5, 6, 7, 8}
- A ∪ D = {1, 2, 3, 4}. Union with the empty set changes nothing.

Memory aid: ∪ looks like a **U**, for **U**nion; it holds everything poured in.

### Intersection: A ∩ B ("A and B")

Only what is in A **and** in B: the elements they share.

- A ∩ B = {3, 4}
- B ∩ C = {6}
- A ∩ C = { }. When two sets share nothing, they are called **disjoint**.
- A ∩ D = { }. Intersection with the empty set is always empty.

Memory aid: ∩ is an upside-down U, like a bridge **over** the shared part.

## Difference: $A \setminus B$ ("A minus B")

Everything in A that is **not** in B. Start with A and cross out anything that appears in B.

- $A \setminus B = \{1, 2\}$
- $B \setminus A = \{5, 6\}$. **Order matters**: $A \setminus B \neq B \setminus A$ in general.
- $C \setminus B = \{7, 8\}$
- $A \setminus D = \{1, 2, 3, 4\}$. Removing nothing changes nothing.

## Complement: $A'$ or $A^c$ ("not A")

Everything in the **universal set U** that is **not** in A. You can only work out a complement if you know U. In fact, $A' = U \setminus A$.

Example: $U = \{1, 2, \ldots, 8\}$. $A = \{1, 2, 3, 4\}$. $A' = \{5, 6, 7, 8\}$.

## Venn diagrams

A **Venn diagram** draws sets as overlapping circles inside a rectangle, which stands for U. With two sets there are four regions:

| Region | Description | Set |
|---|---|---|
| Left part of A only | In A, not in B | $A \setminus B$ |
| The overlap | In both | $A \cap B$ |
| Right part of B only | In B, not in A | $B \setminus A$ |
| Outside both circles | In neither | $(A \cup B)'$ |

And: $A \cup B$ = all three circle regions. $A'$ = B-only region + outside.

## Mock Question 1 solved

The diagram shows A containing a, b, c, e, f and B containing e, f, d, with e and f in the overlap.

So $A = \{a, b, c, e, f\}$, $B = \{d, e, f\}$, $A \cap B = \{e, f\}$, $A \setminus B = \{a, b, c\}$, $B \setminus A = \{d\}$.

Check each option:

- (A) $(A \setminus B) \cap (B \setminus A) = \{a, b, c\} \cap \{d\} = \{\}$. **True.**
- (B) $(B \setminus A) \cup \{a, b, c\} = \{d, a, b, c\}$. That is not B. False.

- (C) A \ (A ∩ B) = {a, b, c}, not {e, f}. False.
- (D) (A ∪ B) ∪ {d} = {a, b, c, d, e, f}, not A. False.

Answer **A**. General fact: A \ B and B \ A never share anything, because one is "only A" and the other is "only B".

## Combining operations: work inside brackets first

**Seminar Task 2.2:** A = {a}, B = {g, h, i, j}, C = {i, j, k, l}, with U = A ∪ B ∪ C.

- B ∩ C = {i, j}
- B ∩ (B ∩ C) = B ∩ {i, j} = {i, j}
- B \ U = { } (everything in B is in U). So A ∪ (B \ U) = A ∪ { } = {a}.

## Data problems (Seminar Task 3.3)

The trick is to **turn each table column into a set** of customer IDs first.

| Set | ShopA customers | ShopB customers |
|---|---|---|
| Electronics | {1, 3, 5} | {4, 5, 7} |
| Clothing | {2, 3, 4} | {3, 5, 6} |
| Groceries | {1, 3, 4} | {3, 6, 7} |

- Electronics in **both** shops: {1, 3, 5} ∩ {4, 5, 7} = **{5}**.
- Clothing in **either** shop: {2, 3, 4} ∪ {3, 5, 6} = **{2, 3, 4, 5, 6}**.
- Groceries in ShopA **but not** ShopB: {1, 3, 4} \ {3, 6, 7} = **{1, 4}**.
- Bought all three in ShopA: only customer 3 has Yes, Yes, Yes. So {3}. Bought at least one thing in ShopB: {3, 4, 5, 6, 7}. Since {3} ⊆ {3, 4, 5, 6, 7}, **yes, it is a subset**.

Key words: **both / and / common** → ∩. **Either / or / at least one** → ∪. **But not / only / without** → difference (A \ B). **Not / outside** → complement.

## Mini tasks 5

U = {5, 6, 7, 10, 11, 12}, A = {5, 6, 10, 12}, B = {5, 7, 11}.

1. A ∪ B
2. A ∩ B

3. A \ B and B \ A
4. A′ and B′
5. A′ ∩ B′
6. Folders: X = {doc1, doc2, doc3, doc4} and Y = {doc3, doc4, doc5, doc6}. Which files must be copied (and where) to make the two folders identical?

## 6. Laws of set algebra and simplifying

These laws let you simplify set expressions without listing elements. Each one is common sense if you picture a Venn diagram.

| Law | Union version | Intersection version |
|---|---|---|
| Commutative (order doesn't matter) | A ∪ B = B ∪ A | A ∩ B = B ∩ A |
| Associative (grouping doesn't matter) | A ∪ (B ∪ C) = (A ∪ B) ∪ C | A ∩ (B ∩ C) = (A ∩ B) ∩ C |
| Distributive | A ∪ (B ∩ C) = (A ∪ B) ∩ (A ∪ C) | A ∩ (B ∪ C) = (A ∩ B) ∪ (A ∩ C) |
| Identity | A ∪ ∅ = A | A ∩ U = A |
| Complement | A ∪ A′ = U | A ∩ A′ = ∅ |
| Idempotent (doing it twice changes nothing) | A ∪ A = A | A ∩ A = A |
| Absorption | A ∪ (A ∩ B) = A | A ∩ (A ∪ B) = A |

Plus the **double negation** law: (A′)′ = A. Not-not-A is A.

### Rules with U and ∅ (Seminar Task 2.3)

Learn these; they come up in "simplify to A, A′, U or { }" questions.

| Expression | Result | Reason |
|---|---|---|
| A ∪ U | U | U already has everything |
| A ∩ U | A | A is inside U |

| Expression | Result | Reason |
|---|---|---|
| A ∪ ∅ | A | Adding nothing |
| A ∩ ∅ | ∅ | Nothing is shared with an empty set |
| A \ U | ∅ | Removing everything |
| A \ ∅ | A | Removing nothing |
| U \ A | A′ | Definition of complement |
| (U \ A) ∪ A | U | A′ ∪ A = U |
| A \ A′ | A | A has nothing in common with A′ |
| U′ | ∅ | Nothing is outside U |
| ∅′ | U | Everything is outside ∅ |

## Method for simplifying

1. Start with the **innermost brackets**.
2. Replace each small piece using the tables above.
3. Keep going outwards until one of A, A′, U or ∅ is left.

## Worked examples (revision Task 13)

**(a) (A ∪ A′) \ (A ∩ A′)**

- A ∪ A′ = U.
- A ∩ A′ = ∅.
- U \ ∅ = **U**.

**(b) ((U \ A) ∪ A)′**

- U \ A = A′.
- A′ ∪ A = U.
- U′ = **∅**.

**(c) (A ∩ (U \ (A ∪ A′))) ∪ (A′ ∩ (A ∪ A′))**

- A ∪ A′ = U, so U \ (A ∪ A′) = U \ U = ∅.
- Left part: A ∩ ∅ = ∅.

- Right part: $A' \cap U = A'$.
- $\varnothing \cup A' = \mathbf{A'}$.

**(d) $((U \setminus (A \setminus A')) \cap (A \cup \varnothing))'$**

- $A \setminus A' = A$.
- $U \setminus A = A'$.
- $A \cup \varnothing = A$.
- $A' \cap A = \varnothing$.
- $\varnothing' = \mathbf{U}$.

### Mini tasks 6

Simplify to A, A′, U or ∅:

1. $(A \cap U) \cup \varnothing$
2. $(A \cup A') \cap A'$
3. $(A')' \cap U$
4. $A \cup (A \cap B)$
5. $(A \setminus U) \cup (U \setminus A)$

## 7. Power sets and Cartesian products

### Power set

The **power set** of S, written **P(S)** (or $2^S$), is the set of **all subsets** of S. It includes the empty set and S itself.

**Example:** S = {a, b}. Its subsets are ∅, {a}, {b}, {a, b}. So P(S) = {∅, {a}, {b}, {a, b}}.

**Cardinality rule:** if |S| = n, then $\mathbf{|P(S)| = 2^n}$. Each element is either in or out of a subset: 2 choices per element, multiplied together.

| \|S\| | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| \|P(S)\| | 1 | 2 | 4 | 8 | 16 | 32 |

**How to list a power set without missing any:** go by size.

**Example:** A = {a, b, c}, so $|P(A)| = 2^3 = 8$.

- Size 0: ∅
- Size 1: {a}, {b}, {c}
- Size 2: {a, b}, {a, c}, {b, c}
- Size 3: {a, b, c}

Total 1 + 3 + 3 + 1 = 8. Correct.

For S = {p, q, r, s}, $|P(S)| = 2^4 = 16$: 1 empty, 4 singles, 6 pairs, 4 triples, 1 whole set.

Note: P(∅) = {∅}, which has **1** element ($2^0 = 1$).

## Cartesian product

The **Cartesian product** A × B is the set of **all ordered pairs (a, b)** with the first item from A and the second from B.

**Example:** A = {1, 2}, B = {a, b}. A × B = {(1, a), (1, b), (2, a), (2, b)}.

**Method:** take each element of A in turn, and pair it with every element of B.

**Order matters!** B × A = {(a, 1), (a, 2), (b, 1), (b, 2)}. That is a different set: (1, a) is not the same pair as (a, 1).

**Cardinality rule:**

```
|A \times B| = |A| \times |B|
```

**Example (revision Task 12):** A = {x, y}, B = {1, 2, 3}.

- A × B = {(x, 1), (x, 2), (x, 3), (y, 1), (y, 2), (y, 3)}. |A × B| = 2 × 3 = 6.
- B × A = {(1, x), (1, y), (2, x), (2, y), (3, x), (3, y)}. Also 6.
- (A × B) \ (B × A): no pair appears in both (every pair in A × B starts with a letter; every pair in B × A starts with a number). So it equals A × B, with cardinality **6**.

## The hard version: when the sets overlap (mock Question 14)

A = {a, b, c, d, e} and B = {d, e, f, g, h, i}. Find |(B × A) \ (A × B)|.

**Step 1. Size of B × A:** |B| × |A| = 6 × 5 = **30** pairs.

**Step 2. Which pairs of B × A are also in A × B?** A pair (p, q) is in both only if p is in A and in B, and q is in A and in B. So both p and q must come from A ∩ B = {d, e}.

The shared pairs are (d, d), (d, e), (e, d), (e, e). That is $|A \cap B|^2 = 2^2 =$ **4**.

**Step 3. Subtract:** 30 − 4 = **26**. Answer **D**.

```
|(B \times A) \setminus (A \times B)| = |A|\,|B| - |A \cap B|^2
```

**Seminar challenge 1.1:** A = {1, ..., 12} (12 elements), B = {8, ..., 22} (15 elements). A ∩ B = {8, 9, 10, 11, 12} (5 elements). $|(A \times B) \setminus (B \times A)| = 12 \times 15 - 5^2 = 180 - 25 =$ **155**.

### Mini tasks 7

1. List P({1, 0}) and give its cardinality.
2. If |S| = 6, what is |P(S)|?
3. A = {a, b, c}, B = {1, 0}. List A × B. What is |A × B|?
4. A = {1, 2, 3}, B = {3, 4}. Find $|(A \times B) \setminus (B \times A)|$.
5. True or false: $\varnothing \in P(S)$ for any set S.

## 8. Intervals

### What an interval is

An **interval** is a continuous stretch of the real number line: **every** real number between two **endpoints**, including all the decimals in between. (1, 3) contains 1.5, 2, 2.0001, 2.9999, and infinitely many more.

Whether the endpoints themselves are included is shown by the type of bracket:

- **Round bracket ( or )** = endpoint **excluded** (like < or >). On a number line: an open, hollow circle.
- **Square bracket [ or ]** = endpoint **included** (like ⩽ or ⩾). On a number line: a filled-in dot.

Memory aid: a square bracket has corners that "grab" the endpoint; a round bracket curves away from it.

### All eight types

| Interval | Name | Set builder form | Contains |
|---|---|---|---|
| (a, b) | Open | {x \| a < x < b} | Between a and b, neither end |
| [a, b] | Closed | {x \| a ⩽ x ⩽ b} | Between a and b, both ends |
| (a, b] | Half-open (left open) | {x \| a < x ⩽ b} | Includes b, not a |

| Interval | Name | Set builder form | Contains |
|---|---|---|---|
| [a, b) | Half-open (right open) | {x \| a ⩽ x < b} | Includes a, not b |
| (a, ∞) | Infinite, open | {x \| x > a} | Everything above a |
| [a, ∞) | Infinite, closed | {x \| x ⩾ a} | a and everything above |
| (−∞, a) | Infinite, open | {x \| x < a} | Everything below a |
| (−∞, a] | Infinite, closed | {x \| x ⩽ a} | a and everything below |

**∞ always gets a round bracket**, because infinity is not a number you can reach or include.

When the question says x is real, the formal version adds "x ∈ ℝ": [2, 9] = {x ∈ ℝ | 2 ⩽ x ⩽ 9}.

## Converting between notations (mock Question 11)

Which set builder notation defines (1, 100]?

- Left bracket "(" → 1 is excluded → **1 < x**.
- Right bracket "]" → 100 is included → **x ⩽ 100**.
- So {x : 1 < x and x ⩽ 100}. Answer **C**.

The other way (lecture): {x | −2 < x < 4} = (−2, 4). {x | 1 ⩽ x < 5} = [1, 5). {x | x ⩽ 0} = (−∞, 0].

## Membership: is a number in the interval?

Check the number against both ends, paying attention to the brackets.

| Question | Answer | Why |
|---|---|---|
| $3.751 \in (2, 5)$? | Yes | Between 2 and 5 |
| $1 \in [1, 4)$? | Yes | [ includes 1 |
| $5 \in (5, 8]$? | No | ( excludes 5 |
| $3 \in (3, 5]$? | No | ( excludes 3 |
| $5 \in (3, 5]$? | Yes | ] includes 5 |
| $0 \in [-2, 2]$? | Yes | Between them |
| $0 \in \{-2, 2\}$? | **No** | Curly braces: a set with only two elements! |

That last row is the most important trap. **$\{-2, 2\}$** is a set with exactly two numbers. **$[-2, 2]$** is an interval with infinitely many numbers.

## Sets versus intervals

| | Set | Interval |
|---|---|---|
| Brackets | { } | ( ) or [ ] |
| Elements | Exactly the ones listed (can be anything) | Every real number in a range |
| Example | {2, 4, 6} has 3 elements | (2, 5) has infinitely many |

## Subsets with intervals (mock Question 9)

Ask: **is every single number in the left thing also in the right thing?** Look for one counterexample: a number that is on the left but not the right. If you find one, the statement is false.

- **$[1, 2] \subseteq \{1, 3\}$?** No. 1.5 is in [1, 2] but not in {1, 3}. (Also 2 isn't in {1, 3}.)
- **$[1, 2] \subseteq [1, 3)$?** Yes. Every number from 1 to 2 is between 1 (included) and 3. **This is the answer, B.**
- **$[1, 2] \subseteq (1, 3)$?** No. 1 is in [1, 2] but (1, 3) excludes 1.
- **$\{1, 3\} \subseteq [1, 2]$?** No. 3 is not in [1, 2].

**A finite set inside an interval:** check each listed element.

- $\{-1, 0, 1\} \subseteq [-1, 1]$? $-1, 0, 1$ are all in it. **True.**
- $\{2, 5\} \subset [2, 5]$? Both are in, and the interval has many more numbers. **True** (proper subset).
- $\{3, 4\} \subset (2, 5)$? 3 and 4 are inside. **True.**
- $\{1, 2, 3\} \subset [1, 2]$? 3 isn't in it. **False.**

**An interval inside a finite set** is almost always false, because the interval contains decimals like 1.5 that the set lacks. (Revision Task 4 option A only counts as true because that question says to treat all elements as natural numbers. Read such instructions carefully.)

**Interval inside interval:** compare the ends.

- $(2, 5) \subseteq [2, 5]$? Yes: $[2, 5]$ has everything $(2, 5)$ has, plus the ends.
- $[2, 5] \subset (1, 5)$? No: 5 is in $[2, 5]$ but not in $(1, 5)$.
- $[2, 5] \subset (2, 6)$? No: 2 is in $[2, 5]$ but not in $(2, 6)$.

## Operations on intervals (mock Question 8)

Union, intersection and difference work on intervals too. Sketch a number line if unsure.

**Mock Q8: $(1, 10) \cap \{10\}$.** The intersection keeps what is in both. The only candidate is 10. Is 10 in $(1, 10)$? No, the round bracket excludes it. So the result is $\varnothing$. Answer **B**.

More examples:

- $[2, 6] \cap (4, 9) = (4, 6]$. The overlap starts just after 4 (excluded by the second) and ends at 6 (included by the first).
- $[1, 5) \cup [5, 8] = [1, 8]$. 5 is included by the second interval, so there is no gap.
- $(1, 5) \cup (5, 8)$ is **not** $(1, 8)$: 5 is missing from both.
- $[1, 10] \setminus \{10\} = [1, 10)$.
- $[0, 5] \setminus (2, 3) = [0, 2] \cup [3, 5]$. Removing the open middle leaves 2 and 3 in.

## Interval transformations (mock Question 13)

Questions of the form **$\{x \in A : (\text{something about } x) \in A\}$** ask: which x in A still land in A after the change?

**Method:**

1. Write the condition as inequalities: lower end $\leqslant$ (expression) $\leqslant$ upper end, keeping each bracket's strictness ($<$ for round, $\leqslant$ for square).

2. Solve for x.
3. Intersect with A itself, because x must also be in A.

**Mock Q13: A = [4, 25). Find {x ∈ A : x² ∈ A}.**

1. $x^2 \in [4, 25)$ means $4 \leqslant x^2 < 25$.
2. x is from A, so $x \geqslant 4$ and positive. Square-root the inequality: $2 \leqslant x < 5$.
3. Combine with $x \in [4, 25)$: x must be at least 4 and below 5. Result **[4, 5)**. Answer **A**.
4. Check: x = 4 gives $16 \in A$ ✓. x = 4.9 gives $24.01 \in A$ ✓. x = 5 gives $25 \notin A$, because 25 is excluded ✗.

Note the trap: answer D is written {4, 5} with curly braces. That is a two-element **set**, not an interval.

**Revision Task 5.1: B = [3, 8]. Find {x ∈ B : 2x ∈ B}.**

- $3 \leqslant 2x \leqslant 8$, so $1.5 \leqslant x \leqslant 4$.
- Combine with $3 \leqslant x \leqslant 8$: **[3, 4]**.

**Revision Task 5.2: C = [4, 20]. Find {x ∈ C : x + 3 ∈ C}.**

- $4 \leqslant x + 3 \leqslant 20$, so $1 \leqslant x \leqslant 17$.
- Combine with $4 \leqslant x \leqslant 20$: **[4, 17]**.

**Seminar Task 6.2(b): A = [2, 9]. Find {x ∈ A : √x ∈ A}.**

- $2 \leqslant \sqrt{x} \leqslant 9$. Square everything: $4 \leqslant x \leqslant 81$.
- Combine with [2, 9]: **[4, 9]**.

**Seminar Task 6.2(a): A = [2, 5]. {x ∈ A : x < 4.5}** → **[2, 4.5)**.

**Challenge 1.2:** same as 6.2(b), plus $x < \sqrt{20}$ (≈ 4.47). Result **[4, √20)**.

## Mini tasks 8

1. Write [−3, 7) in set builder notation.
2. Write {x | x > 2} as an interval.
3. True or false: $9 \in [2, 9)$. And: $2 \in [2, 9)$?
4. True or false: $\{2, 3\} \subseteq (2, 5)$.
5. Find $[0, 10] \cap (5, 12)$.
6. A = [2, 10]. Find $\{x \in A : x^2 \in A\}$.
7. A = [1, 20]. Find $\{x \in A : 4x \in A\}$.

# 9. Sets defined with modular arithmetic

This combines Phase 1 (mod) with set builder notation and set operations. It's Seminar 2 Task 5.

**{x | x ≡ 2 (mod 3), 0 ⩽ x ⩽ 30}** means: every whole number from 0 to 30 that leaves remainder 2 when divided by 3.

**Fast method:** find the **smallest** one, then keep adding the modulus until you pass the upper limit.

- A = {x | x ≡ 2 mod 3, 0 ⩽ x ⩽ 30}: start at 2, add 3 each time. **A = {2, 5, 8, 11, 14, 17, 20, 23, 26, 29}**, |A| = 10.
- B = {x | x ≡ 1 mod 4, 0 ⩽ x ⩽ 30}: start at 1, add 4. **B = {1, 5, 9, 13, 17, 21, 25, 29}**, |B| = 8.
- C = {x | x ≡ 0 mod 5, 0 ⩽ x ⩽ 30}: start at 0, add 5. **C = {0, 5, 10, 15, 20, 25, 30}**, |C| = 7. (0 counts: 0 mod 5 = 0.)

## Operations

**A ∩ B** (in both lists): go through the shorter list B and keep what's also in A: 5 ✓, 17 ✓, 29 ✓. **A ∩ B = {5, 17, 29}**, cardinality **3**.

**A ∪ C** (in either list): A ∩ C = {5, 20}, so |A ∪ C| = 10 + 7 − 2 = **15**.

A ∪ C = {0, 2, 5, 8, 10, 11, 14, 15, 17, 20, 23, 25, 26, 29, 30}.

This uses a handy counting rule called **inclusion–exclusion**:

```
|A \cup B| = |A| + |B| - |A \cap B|
```

You subtract the overlap because those elements were counted twice.

**(B ∪ C) \ A:**

- B ∪ C = {0, 1, 5, 9, 10, 13, 15, 17, 20, 21, 25, 29, 30} (13 elements; 8 + 7 − 2, since 5 and 25 are shared).
- Remove anything in A: 5, 17, 20 and 29 are in A.
- Result **{0, 1, 9, 10, 13, 15, 21, 25, 30}**, cardinality **9**.

## Mini tasks 9

1. List {x | x ≡ 1 (mod 3), 0 ⩽ x ⩽ 15} and give its cardinality.
2. List {x | x ≡ 0 (mod 4), 1 ⩽ x ⩽ 20}.

3. Using your answers to 1 and 2, find their intersection.

## Phase 2 practice questions (exam style)

About 35 minutes. Several are harder than the mock on purpose.

1. What is $|\{1, \{1\}, \{1, 2\}, \{\ \}, \{\ \}\}|$? (A) 3 (B) 4 (C) 5 (D) 6
2. $S = \{a, b, c, d, e\}$. What is $|P(S)|$? (A) 10 (B) 25 (C) 32 (D) 5
3. $U = \{1, \ldots, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$. What is $(A \cup B)'$? (A) $\{7, 8\}$ (B) $\{1, 2, 5, 6\}$ (C) $\{3, 4\}$ (D) $\varnothing$
4. Same sets. What is $(A \setminus B) \cup (B \setminus A)$? (A) $\{3, 4\}$ (B) $\{1, 2, 5, 6\}$ (C) $\{1, 2\}$ (D) $\{1, 2, 3, 4, 5, 6\}$
5. Which statement is true? (A) $\varnothing \in \{1, 2\}$ (B) $\varnothing \subseteq \{1, 2\}$ (C) $\{1\} \in \{1, 2\}$ (D) $\{1, 2\} \subset \{1, 2\}$
6. $A = \{1, 2, 3\}$, $B = \{3, 4\}$. What is $|(A \times B) \setminus (B \times A)|$? (A) 6 (B) 4 (C) 5 (D) 0
7. Which set builder notation defines $[-3, 7)$? (A) $\{x : -3 < x < 7\}$ (B) $\{x : -3 \leqslant x \leqslant 7\}$ (C) $\{x : -3 \leqslant x < 7\}$ (D) $\{x : -3 < x \leqslant 7\}$
8. What is $[2, 6] \cap (4, 9)$? (A) $[4, 6]$ (B) $(4, 6]$ (C) $[2, 9)$ (D) $(4, 6)$
9. Which is true? (A) $[0, 1] \subseteq \{0, 1\}$ (B) $\{0, 1\} \subseteq (0, 1)$ (C) $(0, 1) \subseteq [0, 1]$ (D) $[0, 1] \subset (0, 1)$
10. Which set is $\{x \in \mathbb{Z} : x^2 < 5\}$? (A) $\{0, 1, 2\}$ (B) $\{-2, -1, 0, 1, 2\}$ (C) $\{-2, 2\}$ (D) $(-\sqrt{5}, \sqrt{5})$
11. Which set is $\{x \in \mathbb{Z} : x^2 = -4\}$? (A) $\{-2\}$ (B) $\{-2, 2\}$ (C) $\{2\}$ (D) $\varnothing$
12. $A = [2, 10]$. Which interval is $\{x \in A : x^2 \in A\}$? (A) $[2, \sqrt{10}]$ (B) $[4, 10]$ (C) $[\sqrt{2}, \sqrt{10}]$ (D) $[2, 100]$
13. $A = [1, 9]$. Which interval is $\{x \in A : 3x \in A\}$? (A) $[3, 9]$ (B) $[1, 3]$ (C) $[1/3, 3]$ (D) $[1, 27]$
14. Simplify $(A \cap A') \cup (A \cap U)$. (A) $U$ (B) $\varnothing$ (C) $A$ (D) $A'$
15. What is $(1, 5] \cup \{1\}$? (A) $(1, 5]$ (B) $[1, 5]$ (C) $\{1, 5\}$ (D) $[1, 5)$
16. What is $[0, 10] \setminus (3, 10]$? (A) $[0, 3)$ (B) $[0, 3]$ (C) $(0, 3]$ (D) $[3, 10]$
17. Shop X customers: $\{1, 2, 5, 7\}$. Shop Y customers: $\{2, 3, 7, 8\}$. Which customers bought from **exactly one** shop? (A) $\{2, 7\}$ (B) $\{1, 5\}$ (C) $\{1, 3, 5, 8\}$ (D) $\{1, 2, 3, 5, 7, 8\}$
18. $|A| = 4$, $|B| = 5$ and $|A \cap B| = 2$. What is $|A \cup B|$? (A) 9 (B) 7 (C) 11 (D) 20
19. Which of these is a proper subset of $\{a, b, c\}$? (A) $\{a, b, c\}$ (B) $\{a, d\}$ (C) $\varnothing$ (D) $\{\{a\}\}$
20. $A = \{1, 2\}$. What is $|P(A) \setminus \{\varnothing\}|$? (A) 4 (B) 3 (C) 2 (D) 1
21. Which set is $\{x \in \mathbb{N} : x^2 = 9\}$? (A) $\{-3, 3\}$ (B) $\{3\}$ (C) $\{9\}$ (D) $\varnothing$
22. How many elements does $\{x \mid x \equiv 2 \pmod 5, 0 \leqslant x \leqslant 40\}$ have? (A) 8 (B) 9 (C) 7 (D) 10

# Answers with full explanations

## Mini tasks 1

1. {a, e, i, o, u}
2. {January, March, May, July, August, October, December} (cardinality 7)
3. **True.** Order doesn't matter.
4. $3 \in \{1, 2, 3\}$ is **true**. $\{3\} \in \{1, 2, 3\}$ is **false**: the set {3} is not one of the listed elements.
5. {2, 3, 5, 7}

## Mini tasks 2

1. **4**
2. Pieces: 1, {1}, {1, 2}. **3.**
3. One piece: { }. **1.**
4. Pieces: a, a, b, {a, b}. Remove the duplicate a: **3.**
5. {1, 2} and {2, 1} are the same set (order doesn't matter), so they are duplicates. Distinct: {1, 2}, 3. **2.**

## Mini tasks 3

1. **True** (they're equal, and equal sets are subsets of each other).
2. **False.** A set is never a proper subset of itself.
3. **True.** $\emptyset \subseteq \{a\}$, and $\emptyset \neq \{a\}$.
4. **False.** d is not in {a, b, c}.
5. **False.** $2 \in \{1, 2, 3\}$, but the set {2} isn't listed as an element.

## Mini tasks 4

1. Odd numbers above 4 up to 10: **{5, 7, 9}**.
2. Integers whose square is under 5: −2 (4), −1, 0, 1, 2 (4). ±3 gives 9, too big. **{−2, −1, 0, 1, 2}**.
3. $3^3 = 27$. **{3}**.
4. **{(1, 1), (2, 2), (3, 3), (4, 4)}**, cardinality 4.
5. $\{x \mid x \in \mathbb{N}, x \text{ is even}, x < 10\}$ = **{2, 4, 6, 8}** (4 elements).

## Mini tasks 5

1. $A \cup B = \{5, 6, 7, 10, 11, 12\}$
2. $A \cap B = \{5\}$
3. $A \setminus B = \{6, 10, 12\}$. $B \setminus A = \{7, 11\}$.
4. $A' = U \setminus A = \{7, 11\}$. $B' = \{6, 10, 12\}$.
5. $A' \cap B' = \{7, 11\} \cap \{6, 10, 12\} = \varnothing$. (In general $A' \cap B' = (A \cup B)'$, and here $A \cup B = U$.)
6. $X \setminus Y = \{doc1, doc2\}$ must be copied into Y. $Y \setminus X = \{doc5, doc6\}$ must be copied into X. Then both equal $X \cup Y = \{doc1, ..., doc6\}$.

## Mini tasks 6

1. $A \cap U = A$; $A \cup \varnothing =$ **A**.
2. $A \cup A' = U$; $U \cap A' =$ **A′**.
3. $(A')' = A$; $A \cap U =$ **A**.
4. Absorption law: **A**.
5. $A \setminus U = \varnothing$; $U \setminus A = A'$; $\varnothing \cup A' =$ **A′**.

## Mini tasks 7

1. $P(\{1, 0\}) = \{\varnothing, \{1\}, \{0\}, \{1, 0\}\}$. Cardinality **4** $= 2^2$.
2. $2^6 =$ **64**.
3. $A \times B = \{(a, 1), (a, 0), (b, 1), (b, 0), (c, 1), (c, 0)\}$. $|A \times B| = 3 \times 2 =$ **6**.
4. $|A \times B| = 6$. $A \cap B = \{3\}$, so the only shared pair is $(3, 3)$. $6 - 1 =$ **5**.
5. **True.** The empty set is a subset of every S, so it is an element of P(S).

## Mini tasks 8

1. $\{x \mid -3 \leqslant x < 7\}$
2. $(2, \infty)$
3. $9 \in [2, 9)$ is **false** (round bracket). $2 \in [2, 9)$ is **true**.
4. **False.** 2 is excluded from $(2, 5)$.
5. Overlap from just above 5 to 10, including 10: **(5, 10]**.
6. $2 \leqslant x^2 \leqslant 10$ gives $\sqrt{2} \leqslant x \leqslant \sqrt{10}$ (for positive x). Combine with $x \geqslant 2$: **[2, $\sqrt{10}$]** ($\sqrt{10} \approx 3.16$).
7. $1 \leqslant 4x \leqslant 20$ gives $0.25 \leqslant x \leqslant 5$. Combine with $[1, 20]$: **[1, 5]**.

## Mini tasks 9

1. Start at 1, add 3: **{1, 4, 7, 10, 13}**, cardinality 5.
2. Start at 4 (0 is excluded since the range starts at 1), add 4: **{4, 8, 12, 16, 20}**.
3. Common: **{4}**.

## Phase 2 practice questions

1. **(B)** Pieces: 1, {1}, {1, 2}, { }, { }. The two empty sets are one element. 4 distinct.
2. **(C)** $2^5 = 32$.
3. **(A)** $A \cup B = \{1, \ldots, 6\}$. Outside it in U: {7, 8}.
4. **(B)** $A \setminus B = \{1, 2\}$, $B \setminus A = \{5, 6\}$. Union {1, 2, 5, 6}. This is called the **symmetric difference**: everything in exactly one of the two sets.
5. **(B)** The empty set is a subset of every set. (A) and (C) confuse $\in$ with $\subseteq$. (D): a set is never a proper subset of itself.
6. **(C)** $3 \times 2 = 6$ pairs, minus $|A \cap B|^2 = 1^2 = 1$ shared pair (3, 3). $6 - 1 = 5$.
7. **(C)** "[" means $\leqslant$ and ")" means $<$.
8. **(B)** The overlap starts after 4 (excluded by the round bracket) and ends at 6 (included by the square bracket).
9. **(C)** (0, 1) is [0, 1] minus its ends. (A) fails because of 0.5. (B) fails because 0 is excluded from (0, 1). (D) fails because 0 is in [0, 1] but not in (0, 1).
10. **(B)** Integers are whole numbers, so (D), an interval of reals, is wrong. Negatives count: $(-2)^2 = 4 < 5$.
11. **(D)** No real number squares to a negative.
12. **(A)** $2 \leqslant x^2 \leqslant 10$, so $\sqrt{2} \leqslant x \leqslant \sqrt{10}$. Combined with $x \geqslant 2$: $[2, \sqrt{10}]$. (C) forgets that x must also be in A.
13. **(B)** $1 \leqslant 3x \leqslant 9$, so $1/3 \leqslant x \leqslant 3$. Combined with $x \geqslant 1$: [1, 3].
14. **(C)** $A \cap A' = \varnothing$; $A \cap U = A$; $\varnothing \cup A = A$.
15. **(B)** Adding the number 1 fills in the missing left end: [1, 5].
16. **(B)** Removing (3, 10] takes away everything above 3 but leaves 3 itself. Result [0, 3].
17. **(C)** Union {1, 2, 3, 5, 7, 8} minus intersection {2, 7} = {1, 3, 5, 8}.
18. **(B)** Inclusion–exclusion: $4 + 5 - 2 = 7$.
19. **(C)** $\varnothing \subseteq \{a, b, c\}$ and $\varnothing \neq \{a, b, c\}$. (A) is equal, not proper. (B) has d. (D) contains the set {a}, which is not an element of {a, b, c}.
20. **(B)** $P(A) = \{\varnothing, \{1\}, \{2\}, \{1, 2\}\}$ has 4 elements; removing $\varnothing$ leaves 3.

21. **(B)** −3 is not a natural number. 9 is $x^2$, not x.
22. **(A)** 2, 7, 12, 17, 22, 27, 32, 37 = 8 numbers. The next, 42, is past 40.

**Scoring:** 18 or more out of 22 means you're strong on sets. The most common slip-ups are $\in$ versus $\subseteq$, curly braces versus interval brackets, and forgetting to intersect with A in transformation questions.

When you're ready, ask for **Phase 3: Algorithms, relations, functions and recursion**.